Open Dataset Description Form

Version 1.0 , 20251007

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Motivation for this template

In adherence with the FAIR principle and general data management standards, any openly released data set shall be of the highest quality and usability for people outside the project. This involves a basic explanation of how the data set was obtained, why and how it is useful as well as information on technicalities like data format and legal issues like licensing. In order to allow interesting parties to get in contact with the original data set owners, please provide a contact email address.

This effort is well spent as it will allow people outside of this project to use this data set!

Data identification

Dataset description

This dataset was recorded from a high-volume, high-rate automotive production line for fuel injectors. The primary aim is to support the development of a robust AI model for in-line quality prediction. The data captures various process parameters from different manufacturing stations with the goal of identifying "Not OK" (NOK) parts early, based on their correlation with final End-of-Life (EoL) test measurements.

Data source & provenance

The data originates directly from the production line's sensor and measurement systems. The dataset represents the raw manufacturing data.

Data purposes and uses

The main purpose of this dataset is for a supervised machine learning task: to predict the EOL output value.

Keywords

production; manufacturing; quality prediction; anomaly detection; automotive; injector; machine learning; imbalanced data;

Data engineering

Data set size and file format

The dataset is provided in CSV format. A thousand rows with sixty-seven input features and four output features.

Features of the dataset

The dataset is tabular and contains sixty-seven features corresponding to measurements from the injector production process. These features include timings, pressures, torques, electrical measurements, and other sensor readings from various stations. Due to system-internal naming conventions, many feature names are obfuscated or not immediately descriptive

References related to the dataset

This dataset was generated and used in the context of the SoliDAIR project (Grant Agreement n° 101120276)

Standards and documentation

Data management and organization are guided by Bosch-internal standards

Privacy, ethics and other sensitive information

The dataset contains only machine and process data from a manufacturing line. It does not include any personal, private, or otherwise sensitive information related to individuals.

Anonymization and obfuscation

The dataset has been intentionally anonymized to protect sensitive company and process data. Specifically, the names of the features (process parameters) have been obfuscated and do not reflect their real-world measurement names. This was done to maintain data confidentiality.

Relation to owner and project(s)

Data set original provider

Robert Bosch GmbH

Date of data set release

October 2025

Data set contact information

Data set license

CC BY-NC-ND 4.0

Project involvements

This dataset belongs to the SoliDAIR project (GA n° 101120276), specifically for the use case "UC BOSCH: Robust AI quality prediction in matured, high-rate and high-volume production."

Comments & restrictions

Users of this dataset should be aware of major restriction:

Low Feature Correlation: The provided features have shown very low correlation with the final EoL quality label. Developing a successful predictive model will likely require advanced feature engineering, the use of specialized anomaly detection algorithms, or the generation of synthetic data to augment the minority class.